Y52 Cold Rolling Mill Operation Manual

Chapter 1: Safety & System Architecture

The Y52 Cold Rolling line operates with high-precision hydraulic tension control. Standard operational line speed is calibrated between 120 and 150 m/min.

Section 1.1: Pre-operational Check

1. Inspect oil pressure gauge Y52-OIL-PRS01 to ensure pressure exceeds 4.5 bar.
2. Verify emergency stop circuit continuity across console stations A and B.

Chapter 2: Main Drive Calibration

Section 2.1: AC Inverter Synchronous Speed Setup

The main AC synchronous drive motor (Rated: 400 kW, 380V) requires encoder feedback channel Y52-ENC-FB. Verify line frequency is maintained at 50.0 Hz +/- 0.2 Hz.

Section 2.2: Interlock Verification

When interlocking gate G-01 is opened, drive trip ALM-1002 shall be generated within 150 ms.

Chapter 3: Daily Shutdown Sequence

1. Reduce rolling tension from 45 kN to 10 kN.
2. Stop feed roller motor.
3. Engage mechanical lock pin LP-52.